

Identification and Assessment of Factors Contributing to Traffic Accidents of Taxis in Beijing

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SUMMARY & CONCLUSIONS

In order to study the influencing factors of traffic accidents for taxi drivers in Beijing, and the relationship between the influencing factors and traffic accidents. Based on the research of the related literature and the result of the initial interview, the relationship model of the influencing factors and the traffic accident is established and the corresponding hypothesis is put forward. Through the empirical research, the conclusions are as follows:

- 1) Through literature review and interviews, we found some factors that affect traffic accidents. The influence factors of the traffic accident is divided into three dimensions and 9 factors are organization management, safety culture, natural environment, equipment and facilities, information resources, ideological knowledge, experience and skills, physiological factors, psychological factors.
- 2) Through the study of the three dimensions of the driver (organizational factors, environmental factors and individual factors), the corresponding questionnaire was designed and the validity and reliability of the questionnaire were tested.
- 3) There is a negative correlation between all the factors and traffic accidents, of which the natural environment has the greatest impact on traffic accidents, followed by physiological factors and organizational management.
- 4) We put forward some suggestions for taxi companies from three aspects (establish trade unions within the company, establish a daily network reporting mechanism, adjust the amount of payment), And we will carry out future research on the basis of the proposed measures.

1 INTRODUCTION

Taxi is an important component of urban transportation, which plays a very important role in the development and construction of the city. It can not only facilitate the trip but also expand the employment to increase tax revenues for the city, and has a positive effect on the city's good image [1]. However, due to the taxi in the traffic in the arbitrariness, mobility and more scattered, coupled with the increase in private car road planning unreasonable and other reasons, making the illegal taxi driving and frequent traffic accidents, so the safe hidden trouble should not be ignored[2]. The number of taxis varies from region to region, there are also

significant differences in the incidence of accidents, but overall the taxi accidents remain high [3]. Exploring the factors of traffic accidents caused by taxi drivers is of great significance to safety management. Through this study, we hope to find the main influencing factors that affect the traffic accident, which will help to improve the safety performance of the taxi driver and facilitate the safety management of the taxi company.

Through literature study, different scholars have discussed some factors that influence the job performance, such as the relationship between safety consciousness and job performance, the influence of work pressure on safety behavior and the relationship between organizational factors and job performance [4-6]. For security performance, more from the perspective of safety culture, security atmosphere and leadership decision-making, and the literature research, which is from the whole to study the overall on the impact of security behavior is relatively small. The author will start from the actual work condition of Beijing taxi drivers, using correlation analysis method to study the influence factors of the taxi driver's safety performance, and find out the main influence factors, hoping to provide some guidance for the safe operation of the taxi driver, to improve the safety performance of taxi drivers and ensure the safety of the taxi industry.

2 RESEARCH HYPOTHESIS

The research results show that there are many factors that affect the traffic accidents caused by the taxi drivers, but the accident can't be separated from the three factors: organizational management factors, environmental factors, and individual factors. The author through the initial interview survey found that different traffic accidents involved in the leading factors are different, sometimes is the environmental factors dominate, sometimes is the individual factors dominate. Based on this, this study hopes to identify all possible factors that may affect traffic accidents and identify key factors.

2.1 Organizational Factors

Based on the previous research on the relationship between organizational factors and safety performance, combined with the interviews with taxi drivers, the author divides the organizational factors into organizational management and safety culture. Organizational management

factors refer to organizational system, training mechanism, business goals and so on. Safety cultural factors refers to the safety of enterprises, corporate safety culture [7-8].

A large number of studies have confirmed that organizational management has an impact on job performance [9], while the relationship between organizational management and safety performance is still relatively small. But the organizational factors will directly affect the individual's safety awareness, job satisfaction and other related factors, which will affect the safe operation, therefore it is inferred that the organizational management factors, safety cultural factors and safety accidents have a negative correlation.

2.2 Environmental Factors

Natural environment factors refer to the natural weather conditions faced by taxi drivers in their daily work, including weather conditions such as day and night, as well as wind and snow so on. Facilities and equipment factors include vehicle completeness and interior lighting, vehicle comfort, road conditions etc. Information resources, including some weather forecasts, mobile navigation and other auxiliary resources.

Obviously, the environment has a direct impact on the normal operation of drivers, snow and other weather will affect the driver's line of sight, road congestion will increase the accident rate, and some auxiliary information will interfere with the driver's self-decision-making. Therefore, it is inferred that natural environment, facilities, equipment and information resources are negatively related to traffic accidents.

2.3 Individual Factors

According to the previous research results, combined with the scene of the interview this study will be divided into four aspects of individual factors, including ideological awareness, experience skills, physiological factors, psychological factors [10]. Ideological awareness, including the driver's safety awareness, security attention and so on. Experience skills refers to the driver deal with the sudden problems, road familiarity, driving technical judgment etc. [11]. Physiological factors refer to the driver's compressive capacity, fatigue, health and so on [12]. Psychological factors include job satisfaction and the degree of recognition of drivers [13].

The individual characteristics of the driver are the most direct cause of the accident, so we deduce that the individual factor is negatively correlated with the traffic accident.

2.4 Research Assumptions and Models

Based on the above analysis, the following assumptions are proposed:

Organizational factors, environmental factors and individual factors as the dependent variable, traffic accidents as a result variable.

Organizational factors, environmental factors and individual factors are related to traffic accidents. Specifically, the 2 dimensions of the taxi driver's organizational factors (Organizational management factors, safety culture factors), the 3 dimensions of environmental factors (Natural environment, facilities and equipment, information

resources), and the 4 dimensions of individual factors (Ideological knowledge, experience and skills, physiological factors, psychological factors) are related to traffic accidents one by one.

In summary, the theoretical model of the relationship among taxi drivers, organizational factors, environmental factors, individual factors and traffic accidents is shown in figure 1.



Figure1. The Relationship Model of Traffic Accident and Its Influencing Factors

3 RESEARCH METHODS

3.1 Object of Study

The object of study is the Beijing taxi driver, specifically refers to the taxi driver to obtain the qualifications. Unlicensed taxi, drop taxi, private car drivers are not part of the study.

A total of 90 questionnaires were issued, and 10 initial interviews were conducted to correct the questionnaires and to consider the rationality of the design. The results of the pre-interview were 79.

3.2 Research Tool

The questionnaire on organizational factors, environmental factors, individual factors and traffic accidents was compiled on the basis of the initial interview results and combined with the scale. A total of 41 questions were used in this scale, using the Likert 10 scale, 0 for a total disagreement, and 9 for a complete agreement.

4 DATA STATISTICS AND ANALYSIS

Using SPSS19.0 for data statistics and analysis, including the validity of the questionnaire, reliability test, as well as organizational factors, environmental factors, individual factors and traffic accidents correlation analysis and inspection.

4.1 Sample description

Our research object is Beijing Beiqi taxi group, a total of 79 questionnaires were collected including 75 male and 4 female, they concentrated in the age between 30-50 years of age, years of work concentrated in the 5-20 years, the basic information questionnaire, see table 1.

Table 1 Basic Information on Samples

Item	classification	number	Percentage
age	<30	3	3.8
	30-40	32	40.5
	40-50	37	46.8
	>50	7	8.9
work experience	<5	5	6.3
	5-10	29	36.7
	10-15	19	24.1
	15-20	10	12.7
	>=20	16	20.2

4.2 Reliability and validity test

After analysis by SPSS19.0, the Cronbach's α coefficients of each dimension of the scale were above 0.7, and in general, the coefficient reached 0.7, then passed the reliability test (Table 2).

Table 2 Reliability Analysis

Dimension	Factor	α coefficient	α coefficient
Organizational factors	Organizational management	0.726	0.731
	safety culture	0.734	
	Natural environment	0.876	
environmental factors	facilities and equipment	0.712	0.835
	information resources	0.745	
	Ideological knowledge	0.837	
Individual factors	Experience and skills	0.793	0.801
	psychological	0.836	
	physiological	0.771	

The scale and the Bartlett test were used to analyze the structural validity of the scale, as shown in Table 3, the

KMO value was 0.931, the Bartlett spherical test showed $p < 0.001$, thus rejecting the null hypothesis of the Bartlett spherical test Suitable for factor analysis.

Table 3 Structural Validity Analysis Results

Kaiser-Meyer-Olkin Measure of Sampling Adequacy		0.931
Bartlett's Test of Sphericity	Approx. Chi-Square	392.77
	df	78
	Sig.	.000

4.3 Correlation Analysis

The Pearson correlation coefficient method is used to analyze whether the two consecutive variables are linearly related. The Pearson correlation coefficient (usually expressed by r) is between -1 and 1, and the absolute correlation coefficient is closer to 1, The data points fall tightly on a straight line. Table 4 analyzes the relevance of organizational factors, environmental factors, dimensions, and traffic accidents.

Table 4 Correlation coefficient matrix for each variable

		Traffic accident
Organizational factors	Pearson Correlation	-.427**
	Sig.(2-tailed)	.000
	N	79
environmental factor	Pearson Correlation	-.571**
	Sig.(2-tailed)	.000
	N	79
Individual factors	Pearson Correlation	-.438**
	Sig.(2-tailed)	.000
	N	79
Traffic accident	Pearson Correlation	1
	Sig.(2-tailed)	
	N	79

Note:

*.Correlation is significant at the 0.1 level

**.Correlation is significant at the 0.01 level

***.Correlation is significant at the 0.001 level (2-tailed).

As can be seen from the above table, Beijing taxi company organizational factors, environmental factors, individual factors and traffic accidents were significantly correlated. The correlation coefficient between organization factor and traffic accident is 0.427, the significance level is $p < 0.01$, Environmental factors and traffic accident correlation coefficient r is 0.571, the significance level $p < 0.01$, The correlation coefficient between individual factors and traffic accidents is 0.438, and the significance level is $p < 0.01$. All showed a significant negative correlation.

For the analysis of the correlation between all factors and traffic accidents, see table 5. From the data, we can see that all factors are negatively correlated with traffic accidents.

Through correlation analysis, and the R value as the basis to judge the impact of the strength of the relationship, we can conclude that the influence factors of the traffic accidents can be seen from high to low is the natural environment, physiological factors, organization management, ideological understanding, psychological

factors, information resources, experience and skills, equipment and facilities.

Table 5 Correlation Analysis of Various Factors and Traffic Accidents

Hypothetical relationship	Correlation coefficient
Organizational management → traffic accident	-.532**
safety culture→ traffic accident	-0.213*
Natural environment → traffic accident	-0.679***
facilities and equipment→ traffic accident	-0.174*
information resources→ traffic accident	-0.341**
Ideological knowledge→ traffic accident	-0.474**
Experience and skills→ traffic accident	-0.231**
psychological→ traffic accident	-0.537***
physiological→ traffic accident	-0.437***

5 PROPOSED MEASURES AND FUTURE RESEARCH PROJECTS

5.1 Proposed Measures

The purpose of this study is to identify the factors that affect the taxi drivers' traffic accidents and to assess factors that have a greater impact on traffic accidents. Our research object is Beijing Beiqi Taxi Group, after interviews we found that the company does not care about the workers, apart from the "money". The relationship between taxi drivers and taxi companies is very cold. In addition, the vehicle fuel consumption, maintenance costs and a variety of tickets need the driver himself to bear, so the driver must work more time. The survey found that almost all drivers would work for at least ten hours, and prolonged work not only increased the driver's body load but also affected the driver's family relationship. More importantly, this will lead to high frequency traffic accidents. Through this research situation combined with the data analysis results we propose the following targeted proposals for the Beijing Beiqi taxi group:

- 1) *Establish Trade Unions within the Company:* According to the survey found that the company has no specialized trade union organizations. The responsibilities of the organization include: (1) Regular organization of drivers for health checks; (2) Act as a channel for drivers and company to talk and coordinate their relationships; (3) Balancing labor benefits; (4) Maintain the basic rights and interests of drivers (all kinds of insurance, occupational diseases, accident disputes, etc.)
- 2) *Establish a Daily Network Reporting Mechanism:* The mechanism is mainly to facilitate the company to manage the daily work of the driver. The main way is to establish a daily employee information management module on the company's internal website, the login of the module requires the driver's license number and login password. The contents include the day's oil consumption, the day of the vehicle situation, the

income situation, the main location of the car, weather conditions, road conditions, and accident reporting situation. The information is filled in as a daily attendance situation. Those who have a monthly attendance rate lower than the required requirements need to pay a certain fine, of course, the driver who full attendance will have perfect attendance award.

- 3) *Adjust the Amount of Payment:* On the issue of the daily payment of the driver should be used "two-handed" mechanism. On the one hand the company need to budget out of all aspects of the operating costs of the taxi. Such as the cost of car replacement costs, the cost of staff training, taxi facilities and equipment costs, etc, and ultimately calculate the operating costs of each taxi; On the other hand, according to the urban transport demand, the total capacity, roughly calculate the reasonable monthly income of each taxi C_0 . Combined with the company operating costs and the driver's income in two areas to get a balance coefficient ε ($0 \leq \varepsilon \leq 1$). In addition, counting the actual monthly income of the driver C_1 . C as the actual amount of money the driver needs to pay ($C = C_1 * \varepsilon$). Drivers who do not reach the average income of C_0 are required to pay a fine, and the fines are used to reward the driver with superior safety performance.

5.2 Future Research Plan

We will still choose Beijing Beiqi Taxi Group as the object of study. The research plan for the future is as follows:

- 1) We will choose a driver's team of the company as the test object, and on the basis of our proposed measures to rectify the driver's team.
- 2) We will try to get in touch with the government and help us do some research work.
- 3) After a period of practice, we will re-interview and distribute questionnaires to the drivers of the team, and re-analyze the data collected, and compare the results of previous data analysis.

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